

CoRoT-2b

R-2368

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_100724 · created 2026-08-20T00:06:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455401.85674 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.046
Transit depth (Rp/Rs)^2	2.35 +/- 1.42 [%]
Orbital Inclination (inc)	89.85 +/- 1.78
Airmass coefficient 1 (a1)	1.269 +/- 0.029
Airmass coefficient 2 (a2)	-0.14 +/- 0.024
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	None
Optimal Aperture	2.32
Optimal Annulus	9.24
Transit Duration (day)	0.0847 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.153 $\pm$ 0.046 vs 0.1667 (deviation 0.23 σ)
Duration fit vs published	0.0847 d vs 0.095 d (deviation 3.22 σ)
Mid-time O-C	-2.5 min
Depth SNR	2.6
Residual scatter	2.29 %

Light curve

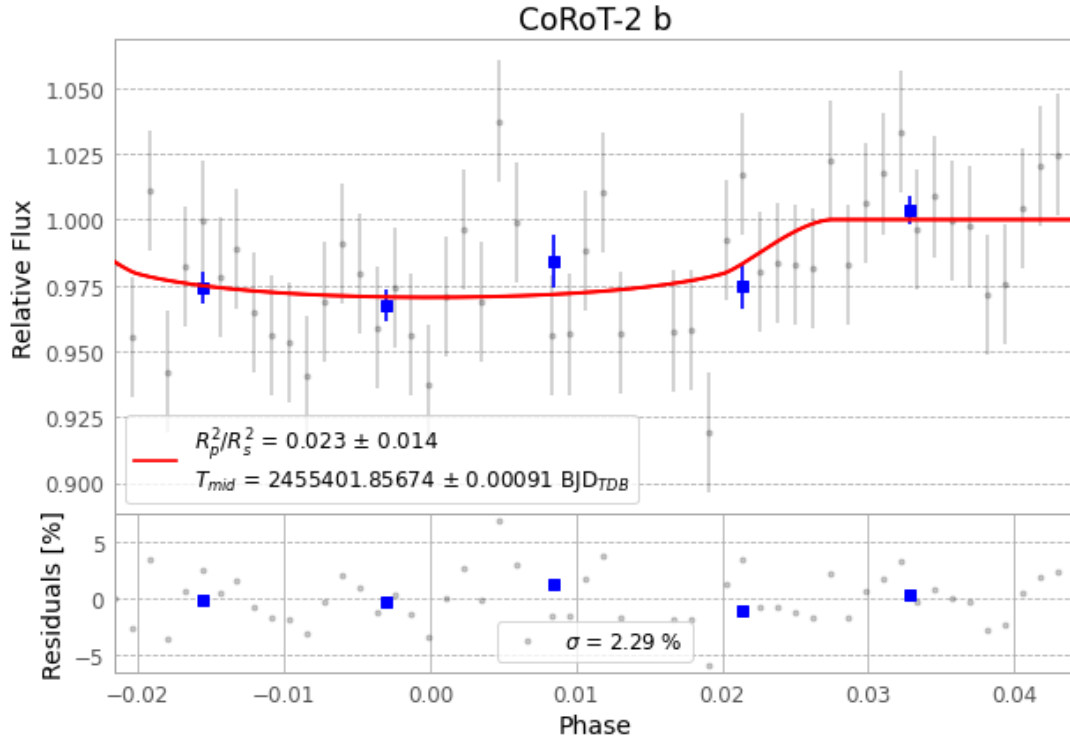


Figure 1 — FinalLightCurve_CoRoT-2 b_2010-07-24.png (SHA-256 Oddadf1ee5c1bbd1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3d139d542e539675...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

54 FITS frames (35 MB), CoRoT-2100724073014.FITS → CoRoT-2100724101514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-100724.log (010c09b246e3...), FinalLightCurve_CoRoT-2 b_2010-07-24.png (Oddadf1ee5c1...), FinalLightCurve_CoRoT-2 b_2010-07-24.csv (0946425fbccb...)

Compute resources

Wall time 405.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:06Z.